BRING-UP REPORT (SIMULATED BENCH)

hifi-usb

Run 20261014-051551-sim: passed, 17 of 17 steps passed.

17	0	0	4
steps passed	steps failed	steps not reached	human confirmations

Run

ITEM	VALUE
Board	hifi-usb
Bench	sim
Started	2026-10-14T05:15:51Z
Finished	2026-10-14T05:15:51Z
Procedure	aec51440615eb619 (generated 2026-10-14T05:15:50Z)
Firmware	0000000000000000
netlist	9954f200149f13ca
constraints	df4c042560edbef0
bom	b184327335ea5085
requirements	1a3d92a00ab4fb98
manifest	5e9fa4d75dbebc5b
Simulated faults	none
Seed	1

Stages

STAGE	STEPS	PASS	FAIL	NOT RUN
usb	8	8	0	0
pd	9	9	0	0

Not in this procedure: visual, unpowered (a USB-first board is plugged straight into a host; its port's current limit stands in for the bench supply).

Measurements

STEP	WHAT	LIMITS	VALUE	VERDICT
usb.02	U2 CH340C enumerates as 1a86:7523		/dev/sim-ch340	pass
usb.03	esptool over the board's USB			pass
usb.04	reset into the app, boot log to the banner		hifi-fw sim	pass
usb.05	rails on USB power	vbus_v 4.4 .. 5.25 V; v3v3_v 3.135 .. 3.465 V	vbus_v=5.017, v3v3_v=3.288	pass
usb.06	1000 Hz at -20 dBFS read back on TAP_DAC_L	freq_hz 995 .. 1005 Hz; vrms 0.18716 .. 0.23562 Vrms; thdn_db <= -85 dB	freq_hz=999.98, vrms=0.20823, thdn_db=-93	pass
usb.07	1000 Hz at -20 dBFS read back on TAP_DAC_R	freq_hz 995 .. 1005 Hz; vrms 0.18716 .. 0.23562 Vrms; thdn_db <= -85 dB	freq_hz=999.84, vrms=0.2111, thdn_db=-94.2	pass
usb.08	on Wi-Fi: the console's TCP address	port 1 .. 65535; rssi_dbm -75 .. 0 dBm	port=3333, rssi_dbm=-62	pass
pd.02	console over Wi-Fi (TCP)			pass
pd.03	PD contract 28 V or 20 V	contract_v 26.6 .. 29.4 V or 19 .. 21 V	contract_v=28.07	pass
pd.04	amp rail VAMP	vamp_v 26.6 .. 29.4 V or 19 .. 21 V	vamp_v=28.075	pass
pd.06	1000 Hz at -20 dBFS read back on TAP_SPK_L	freq_hz 995 .. 1005 Hz; vrms 0.74511 .. 0.93804 Vrms; thdn_db <= -65 dB	freq_hz=1000.12, vrms=0.8417, thdn_db=-72.1	pass

STEP	WHAT	LIMITS	VALUE	VERDICT
pd.07	1000 Hz at -20 dBFS read back on TAP_SPK_R	freq_hz 995 .. 1005 Hz; vrms 0.74511 .. 0.93804 Vrms; thdn_db <= -65 dB	freq_hz=999.93, vrms=0.83866, thdn_db=-71.4	pass
pd.08	1000 Hz at -20 dBFS read back on TAP_SUB	freq_hz 995 .. 1005 Hz; vrms 0.18716 .. 0.23562 Vrms; thdn_db <= -80 dB	freq_hz=999.8, vrms=0.20656, thdn_db=-88.6	pass

Human steps

A simulated run confirms its own human steps as `sim`.

STEP	ACTION	BY	AT	RESULT
usb.01	Plug the board's USB-C (J1) into this host with a USB data cable (not charge-only). Nothing else on the board: no charger, no speakers. It powers up from the host's 5 V now.	sim	2026-10-14T05:15:51Z	pass
pd.01	Unplug the USB cable from the host and plug a USB-C PD charger with EPR (28 V) that offers 28 V or 20 V into the same USB-C (J1). The board restarts on the charger and the console carries on over Wi-Fi.	sim	2026-10-14T05:15:51Z	pass
pd.05	Connect the speakers to the speaker outputs and the sub to the sub out. The next steps play a 1000 Hz tone at -20 dBFS through each output in turn.	sim	2026-10-14T05:15:51Z	pass
pd.09	Each speaker and the sub played the tone in turn, cleanly, with no buzz or hum between tones.	sim	2026-10-14T05:15:51Z	pass